

trace inequality

OPG-3031

L1 - TRACTABLE

PARTIALLY SOLVED

Algebra

ORIGINAL PROBLEM STATEMENT

Let A, B be positive semidefinite, by Jensen's inequality, it is easy to see $[tr(A^s + B^s)]^{\frac{1}{s}} \leq [tr(A^r + B^r)]^{\frac{1}{r}}$, whenever $s > r > 0$.

What about the $tr(A^s + B^s)^{\frac{1}{s}} \leq tr(A^r + B^r)^{\frac{1}{r}}$, is it still valid?